Database Project Documentation – Bookstore Management System

1. Introduction

This project consists of the development of a **relational database for a bookstore**, created using **Microsoft Access**.
The goal of the project is to manage the main activities of a bookstore, such as **books, customers, suppliers, sales, purchases, inventory and data analysis**, through a structured and user-friendly system.

The project follows the main steps of database design:

- conceptual design (ER model),
- logical design (tables and relationships),
- implementation of queries,
- creation of a graphical user interface (forms and buttons).

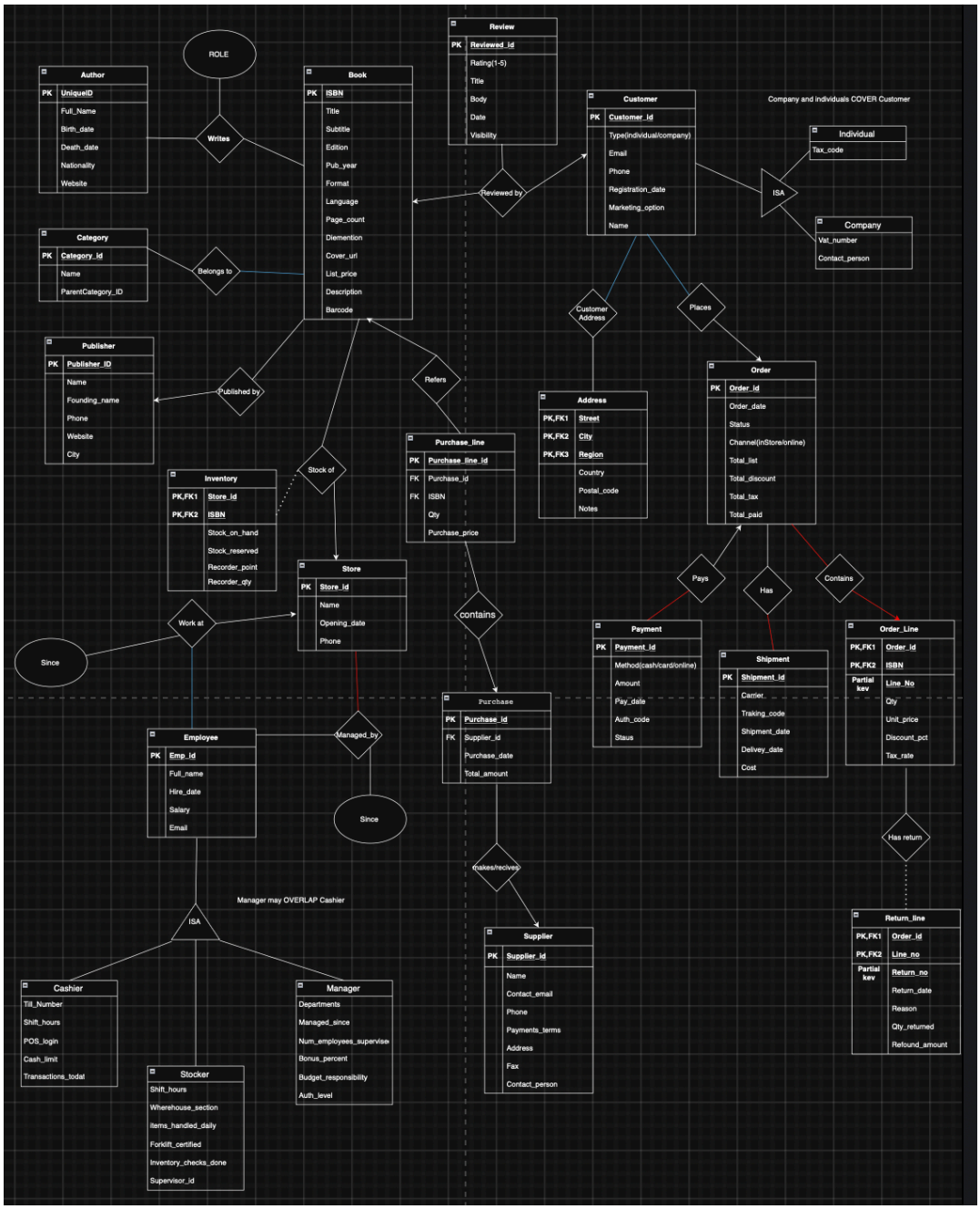
2. Conceptual Design (ER Model)

The first step was the creation of an **Entity-Relationship (ER) model**, used to represent the structure of the data and the relationships between entities.

The main entities are:

- **Book**
- **Customer**
- **Supplier**
- **Publisher**
- **Category**
- **Order** and **Order_line**
- **Purchase** and **Purchase_line**
- **Inventory**
- **Store**

Each entity has a primary key, and relationships between entities are represented using foreign keys.
Many-to-many relationships are resolved using intermediate tables (for example, `Order_line` and `Purchase_line`).
The ER model ensures data consistency and avoids redundancy.



3. Database Implementation in Microsoft Access

The ER model was implemented in **Microsoft Access** by creating the corresponding tables. Relationships between tables were defined using the **Relationships** window in Access.

Key aspects of the implementation:

- one-to-many relationships were correctly defined,
- **referential integrity** was enforced,
- primary keys use Autonumber fields,
- foreign keys ensure correct data linkage.

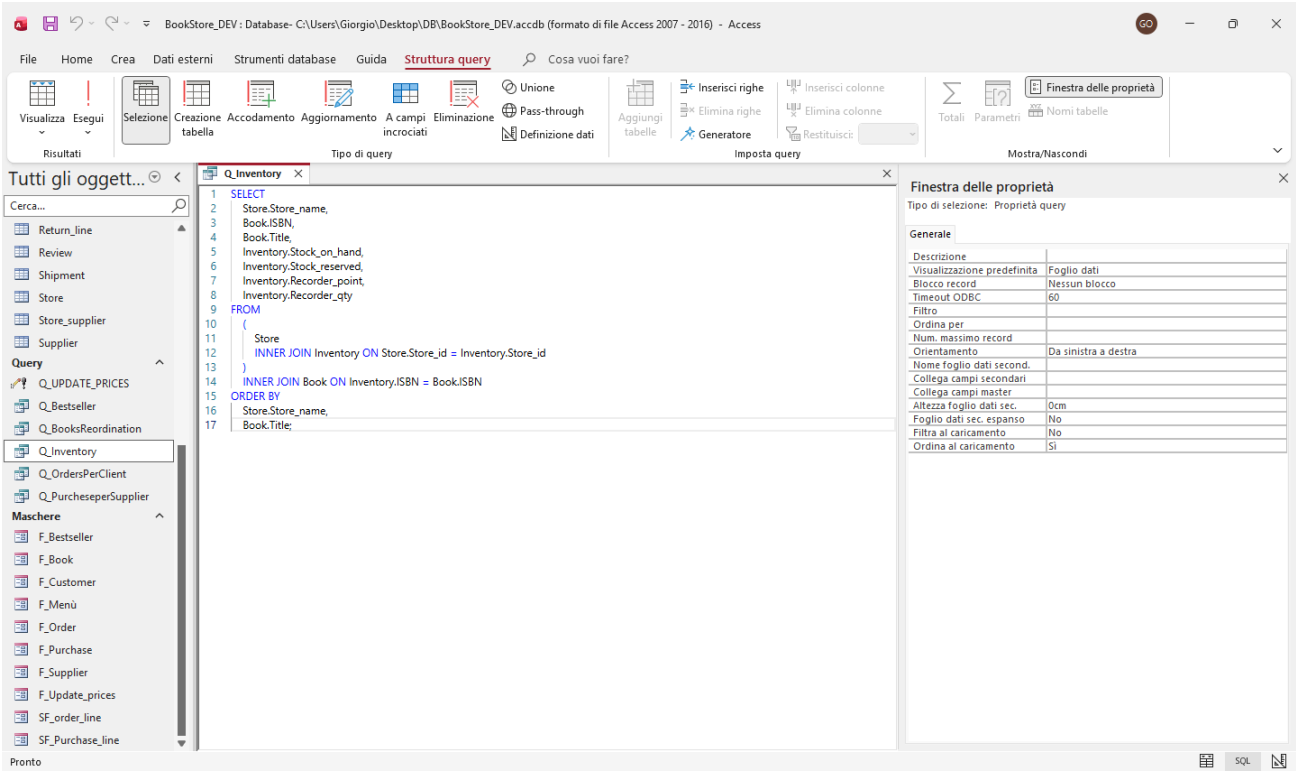
This structure guarantees data accuracy and prevents invalid records.

4. Queries

Several queries were created to analyze and manage the data:

- **Orders per customer**: shows customer sales with books and quantities.
- **Purchases per supplier**: analyzes purchases made from suppliers.
- **Bestsellers by period**: shows the most sold books in a selected date range.
- **Inventory**: displays stock information for books.
- **Books to reorder**: identifies books that need restocking.
- **Price update**: updates book prices by publisher.

The queries were implemented using **SQL and query design tools**, without using advanced features not covered during lessons.



An example of query

5. User Interface (Forms)

To make the database easy to use for non-technical users, several **forms** were created:

- **F_Book** – book management
- **F_Customer** – customer management
- **F_Supplier** – supplier management
- **F_Order** with subform **Order_line** – sales management
- **F_Purchase** with subform **Purchase_line** – purchase management
- **F_BestsellerPeriodo** – bestseller analysis by date
- **F_Update_prices** – book price update by publisher

Combo boxes are used to select related data (for example, customer names or publishers), hiding numeric IDs from the user.

6. Buttons and Navigation

A **main menu form (F_MenuPrincipale)** was created to guide the user through the application using buttons.

The menu allows access to:

- book, customer, sales and purchase management,
- bestseller analysis,
- price update function,
- exit from the application.

This design makes the database similar to a real business application.

7. Developer Version and User Version

Two versions of the database were created:

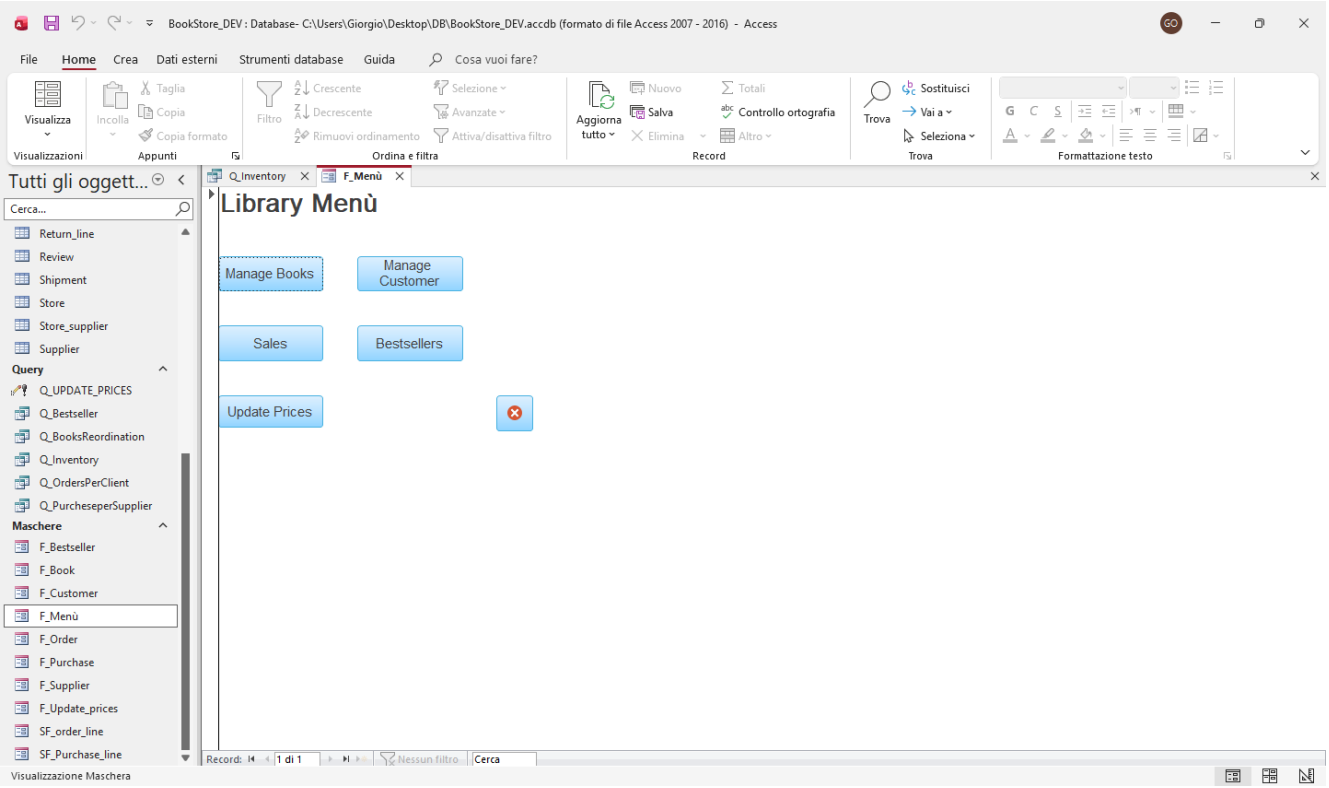
- **Developer version (DEV)**: full access to tables, queries and structure.
- **User version (USER)**: limited access, designed for bookstore employees.

In the User version:

- the navigation pane is hidden,
- special Access keys are disabled,

- users interact only through forms and buttons.

This separation simulates a real working environment.



The Main Menù

8. Conclusion

The project demonstrates the correct design and implementation of a **relational database for a bookstore**.
It applies database design principles, data integrity rules and user-friendly interfaces.

The system supports both daily operations and data analysis, showing how Microsoft Access can be used to create a complete database application.